Contest Problem Set 12227

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Sprint Round 12227

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1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. Annalise purchased three packages of gum at the store. Each package of gum cost \$2. Annalise paid the cashier for the gum with a \$10 bill. How many dollars did Annalise receive in return from the cashier?

(A) \$6 (B) \$2 (C) \$8 (D) \$5 (E) \$4

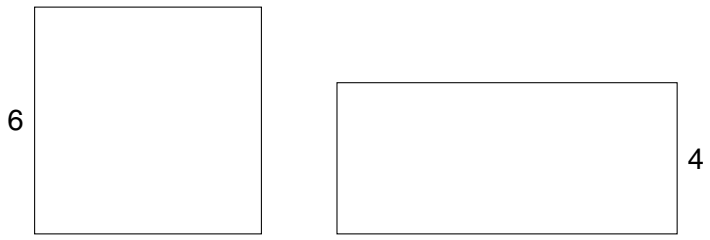
2. Which of the following sums is odd?

(A) $14 + 10 + 18$ (B) $13 + 16 + 19$ (C) $12 + 17 + 18$ (D) $11 + 15 + 16$ (E) $14 + 19 + 15$

3. Kaitlyn is purchasing fruit for a large picnic. At the store, Kaitlyn bought two bags of oranges, and each bag held one dozen oranges. She bought four bunches of bananas. Three of the bunches had seven bananas and one bunch had six bananas. She also purchased two sacks of apples, and each sack held ten apples. Altogether, how many oranges, bananas, and apples did Kaitlyn purchase for her picnic?

(A) 62 (B) 70 (C) 81 (D) 71 (E) 61

4. A square and a rectangle have the same area. The side length of the square is 6, and the width of the rectangle is 4. What is the perimeter of the rectangle?



(A) 20 (B) 22 (C) 28 (D) 26 (E) 24

5. Linus chose a positive whole number. He tripled the number, and then subtracted five from that result. Then he doubled that result. The result of Linus' operations was 44. What number did Linus choose?

(A) 12 (B) 9 (C) 10 (D) 11 (E) 13

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6. Evan calculated $222 \times 17 + 333 \times 17 + 444 \times 17$ and then summed the digits of the result. What was Evan's sum?

(A) 36 (B) 27 (C) 29 (D) 33 (E) 31

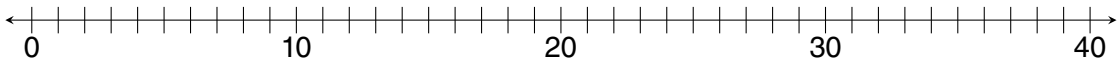
7. April 1st of the year 2022 was a Friday. The month of April has 30 days. How many days in the month of April in the year 2022 were either Saturdays or Sundays?

(A) 8 (B) 7 (C) 10 (D) 9 (E) 6

8. How many positive whole numbers less than 200 are evenly divisible by 9 but not evenly divisible by 6?

(A) 14 (B) 10 (C) 22 (D) 11 (E) 12

9. Chip chose a number on the number line. The distance between Chip's number and 30 was twice the distance between Chip's number and 21. What is the sum of all possible numbers that Chip chose?



(A) 39 (B) 36 (C) 12 (D) 27 (E) 24

10. The sum of three numbers is $\frac{1}{2}$. Two of the numbers are $\frac{1}{5}$ and $\frac{1}{4}$. What is the value of the second greatest of the three numbers minus the least of the three numbers?

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{3}{20}$ (D) $\frac{1}{10}$ (E) $\frac{1}{20}$

11. George is arranging his four Transformers toys in a row on a shelf. He wants to place his Optimus Prime toy next to his Bumblebee toy, but he does not want to place his Megatron toy next to his Bumblebee toy. He does not care where his Starscream toy is placed. How many different ways can George arrange his Transformers toys?

(A) 6 (B) 2 (C) 9 (D) 8 (E) 4

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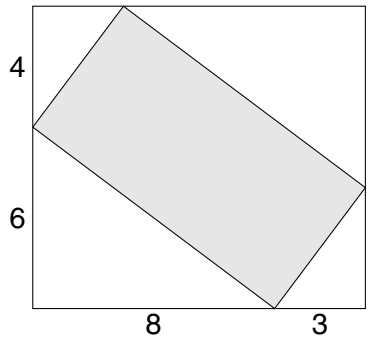
12. The corners of a smaller shaded rectangle lie on the sides of a larger rectangle, as shown below. One side of the larger rectangle is divided into lengths of 4 and 6 by one corner of the shaded rectangle, and another side of the larger rectangle is divided into lengths of 8 and 3 by another corner of the shaded rectangle. What is the area of the shaded rectangle?



(A) 56 (B) 48 (C) 50 (D) 60 (E) 54

13. At Highville Park Middle School, the ratio of fifth grade students to sixth grade students on the Math Team was 5 : 6. After Spring Break, 4 more fifth grade students and 1 more sixth grade student joined the team, and the new ratio of fifth grade students to sixth grade students on the Math Team was 1 : 1. After the additional students joined, how many total fifth and sixth grade students were on the Math Team?

(A) 38 (B) 34 (C) 32 (D) 40 (E) 36

14. Andrew, Brandon, and Clint are friends competing against one another in swimming, running, and cycling races. Andrew finished in first place in the swimming race, and Brandon finished in second place in the running race. None of the three friends finished in the same place in more than one race. Using the letter A to represent Andrew, the letter B to represent Brandon, and the letter C to represent Clint, in what order, from first through third, did the three friends finish in the cycling race?

(A) A, B, C (B) C, B, A (C) C, A, B (D) B, C, A (E) B, A, C

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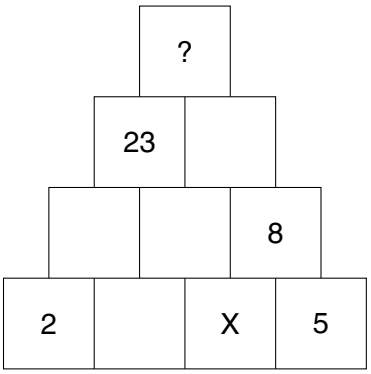
15. In the number pyramid below, the number in a square is the sum of the numbers in the two squares in the row below and next to the square. For example, 8 is the sum of 5 and the number in the square marked *X*. What number belongs in the top square of the number pyramid?

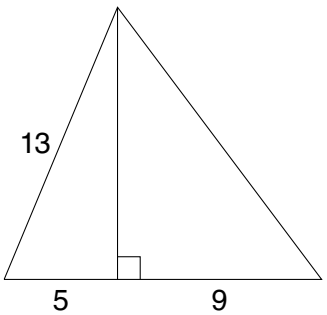


(A) 43 (B) 45 (C) 48 (D) 39 (E) 46

16. Emily is a star libero on her volleyball team. In volleyball, a match has three games. In two matches of a recent tournament, Emily averaged 11 digs per game in the first match, and in the three games of the second match, Emily had 9 digs, 14 digs, and 16 digs. How many digs per game did Emily average for the two matches?

(A) 13.5 (B) 12 (C) 11.5 (D) 12.5 (E) 13

17. In the large triangle below, one side has length 13. A segment drawn from one vertex to another side is perpendicular to that side, and partitions that side into segments of length 5 and 9. What is the perimeter of the large triangle?



(A) 42 (B) 41 (C) 44 (D) 43 (E) 48

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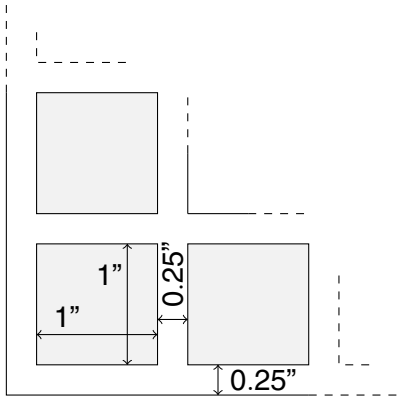
18. How many prime numbers are greater than 150 and less than 200?

(A) 9 (B) 11 (C) 12 (D) 13 (E) 10

19. Yesterday, Devan walked one mile of his trip from home to school and then, because he was running late, he ran the last quarter mile to school. Devan runs twice as fast as he walks, and the running portion of his trip lasted $2\frac{1}{2}$ minutes. How many total minutes was Devan's trip to school yesterday?

(A) 20 (B) 18 (C) $17\frac{1}{2}$ (D) 25 (E) $22\frac{1}{2}$

20. An artist is constructing a rectangular mosaic. The artist is using 1 inch by 1 inch square tiles, with a border of 0.25 inches around each tile, including around the outside of the outer edge of tiles. One corner of the mosaic is shown below. If the rectangular mosaic is 27 tiles wide and 17 tiles high, then what is the total area covered by the mosaic, in square inches?



(A) $717\frac{3}{16}$ (B) 731 (C) 459 (D) $725\frac{5}{8}$ (E) $722\frac{1}{2}$

21. If $a\diamondsuit b = 4ab + 2a + 2b + 1$, then what is the value of $126\diamondsuit 128$?

(A) 65025 (B) 65010 (C) 65000 (D) 65021 (E) 65001

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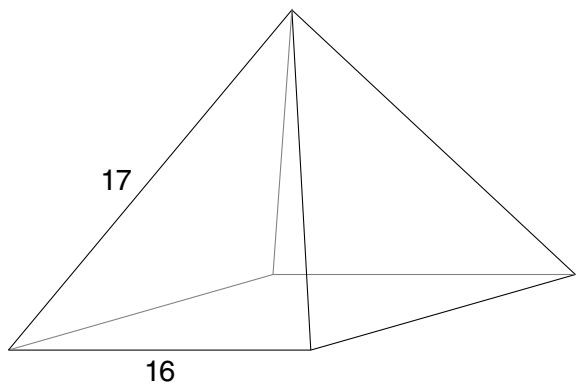
22. A pyramid has a square base of edge length 16, and all four edges from the base to the apex, or peak, of the pyramid have length 17. What is the total surface area of the pyramid?



(A) 756 (B) 736 (C) 784 (D) 748 (E) 800

23. How many different arrangements of the letters *KENOBI* do not have the *K* next to the *N*, the *K* next to the *B*, or the *N* next to the *B*?

(A) 120 (B) 108 (C) 72 (D) 36 (E) 144

24. Huey chose a positive whole number in base 10. He then converted the number to base 9 and wrote it on a piece of paper. Louie saw the number on the piece of paper, and he read the digits of the number correctly, but he mistakenly read base 9 as base 7. He then converted what he thought was a base 7 number to base 10. Louie's result was 512. What number did Huey choose in base 10?

(A) 729 (B) 1331 (C) 1000 (D) 1728 (E) 512

25. How many perfect squares evenly divide 10!?

(A) 30 (B) 36 (C) 48 (D) 24 (E) 40

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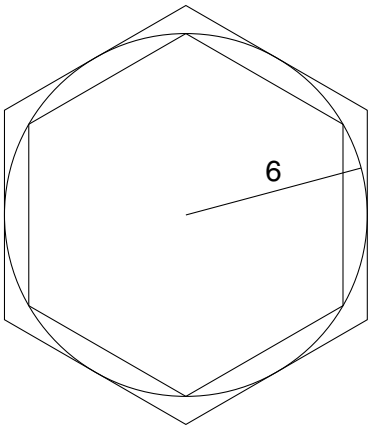
26. Dan is running clockwise around a track at a constant speed. His son Dave is walking slowly around the same track at a constant speed, but in a counterclockwise direction. Dan completes one lap around the track every 90 seconds, and he passes Dave every 65 seconds. How many seconds does it take Dave to complete one lap around the track?

(A) 216 (B) 231 (C) 221 (D) 234 (E) 240

27. A circle of radius 6 is inscribed in a regular hexagon. The same circle is also circumscribed about a regular hexagon. What is the area of the region within exactly one of the hexagons?



(A) $12 + 12\sqrt{3}$ (B) $42 - 6\sqrt{3}$ (C) $24\sqrt{3}$ (D) $18\sqrt{3}$ (E) $54 - 12\sqrt{3}$

28. Cassidy wrote down a list of seven positive whole numbers that formed an arithmetic sequence. The first number in Cassidy's list was 3, and the average of the last four numbers in Cassidy's list was three times the average of the first three numbers of Cassidy's list. What was the sum of the fourth and fifth numbers of Cassidy's list?

(A) 34 (B) 29 (C) 39 (D) 42 (E) 31

29. What is the value of $\sqrt{1.5 + 0.5\sqrt{5}} - 0.5\sqrt{5} + 0.5$?

(A) $\frac{5}{2}$ (B) 1 (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{2}{5}$ (E) $\sqrt{5} - 1$

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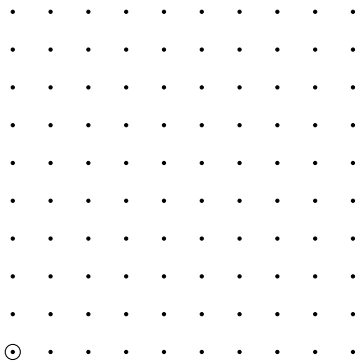
30. The figure below shows 100 dots arranged into ten rows and ten columns. In each row, the dots are one unit apart horizontally, and in each column, the dots are one unit apart vertically. How many different straight lines pass through the circled dot in the lower left corner of the figure and at least one other dot in the figure?



(A) 65 (B) 57 (C) 51 (D) 99 (E) 64

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1. $221 - 22 =$ _____.
2. $569 + 242 =$ _____.
3. $9 + 11 \times 9 =$ _____.
4. The units digit of 72801 is _____.
5. $602 - 396 =$ _____.
6. The remainder of $117 \div 5$ is _____.
7. $24 + 53 + 56 + 26 =$ _____.
8. $7 \times 17 =$ _____.
9. The quotient of 92 divided by 4 is _____.
10. (estimate) $1506 + 2761 + 3263 =$ _____.
11. $94 \times 11 =$ _____.
12. $16^2 =$ _____.
13. The remainder of $327 \div 8$ is _____.
14. $531 - 229 - 212 =$ _____.
15. $56 \times 15 =$ _____.
16. $58 + 87 + 145 =$ _____.
17. $19 \times 5 \times 18 =$ _____.
18. *MCCCLI* in Arabic numerals is _____.
19. $1.92 \div 0.024 =$ _____.
20. (estimate) $199 \times 329 =$ _____.
21. The GCD of 28 and 52 is _____.
22. $57 \times 25 =$ _____.
23. The lesser of $\frac{7}{12}$ and $\frac{3}{5}$ is _____ (fraction).
24. $15 \times 77 \div 35 =$ _____.
25. $41 + 34 + 27 + 20 + 13 + 6 =$ _____.
26. $\frac{3}{35} + \frac{1}{7} + \frac{1}{5} =$ _____ (fraction).
27. $918 \div 34 =$ _____.
28. If 1 mile is equal to 1760 yards, then 176 yards is equal to _____ mile (fraction).
29. $31^2 =$ _____.
30. (estimate) $99 \times 9 \times 99 =$ _____.
31. $46 \times 44 =$ _____.
32. The area of a rectangle with width 9 and perimeter 42 is _____.
33. $\sqrt{784} =$ _____.
34. 2 days is _____ minutes.
35. $\frac{31}{50} =$ _____ %.
36. The LCM of 28 and 52 is _____.
37. $95^2 =$ _____.
38. The number of even whole numbers between 41 and 191 is _____.
39. $99 \times 41 =$ _____.
40. (estimate) $444 \times 629 =$ _____.

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41. $39^2 - 14^2 =$ _____.

42. The fifteenth term in the arithmetic sequence $9, 20, 31, \ldots$ is _____.

43. $87 \times 27 =$ _____.

44. $18 \times 7 + 26 \times 9 =$ _____.

45. The remainder of $2986 \div 11$ is _____.

46. $101 \times 51 =$ _____.

47. 144_9 in base 10 is _____.

48. The perimeter of an isosceles triangle with sides of length 2 and 7 is _____.

49. $4^5 =$ _____.

50. (estimate) $428571 \times 9 =$ _____.

51. $105 \times 117 =$ _____.

52. The perimeter of a right triangle with a leg of length 9 and a hypotenuse of length 41 is ____.

53. 65_8 in base 2 is _____ $_2$.

54. $64 \times 111 =$ _____.

55. $36\frac{4}{11}\%$ of 143 is _____.

56. $\sqrt{6889} =$ _____.

57. The mean of the list $18, 15, 32, 11, 29$ is _____.

58. $97 \times 77 =$ _____.

59. The measure of an interior angle in a regular dodecagon is _____ °.

60. (estimate) $938772 \div 229 =$ _____.

61. $15 \times 375 =$ _____.

62. $11\frac{2}{3} \times 1\frac{2}{3} =$ _____ (mixed number).

63. The number of different two-letter arrangements of the letters *HOWDY* is _____.

64. $0.5\overline{6} =$ _____ (fraction).

65. $12^3 =$ _____.

66. If $v = 16$, then $v^2 + 28v + 196 =$ _____.

67. $16 \times 42 \times 13 =$ _____.

68. The number 74 in base 5 is _____ $_5$.

69. $28\frac{4}{7}\% =$ _____ (fraction).

70. (estimate) $7.07^6 =$ _____.

71. $1029 \times 1031 =$ _____.

72. The sum of the terms of the infinite geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \ldots$ is _____.

73. $44 \times 15 \times 64 =$ _____.

74. The number of subsets of the set $\{Q, U, A, L, I, F, Y\}$ is _____.

75. $729^{\frac{5}{6}} =$ _____.

76. The number of positive whole number divisors of 154 is _____.

77. $\sqrt{32} \times \sqrt{162} =$ _____.

78. If $4^{x-1} = 9$, then $2^{x+1} =$ _____.

79. The volume of a cube with surface area 726 is _____.

80. (estimate) $(2.022 \times 20 \times 2.2)^2 =$ _____.

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Problems 1 & 2

1. Maddy purchased a box of Mint Thins cookies from her friend Claudia, who was selling them as a fundraiser. Inside the box of Mint Thins, the cookies were arranged in two columns. In each column, there were two packets of cookies. In each packet of cookies, there were nine cookies. If Maddy ate three cookies per day, then how many days did it take her to eat all of the cookies in her box of Mint Thins?

1.

2. Thomas chose a three-digit positive whole number. The number Thomas chose was divisible by both 7 and 11, but it had a remainder of 2 when divided by 13. What number did Thomas choose?

2.

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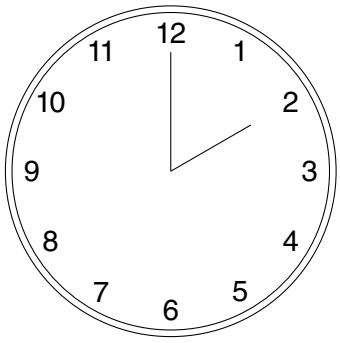
Problems 3 & 4

3. Captain Marbles the kitten weighed 4 pounds at the beginning of January. In the month of January, Captain Marbles gained 1 pound of weight. For each of the three months after January, Captain Marbles gained $\frac{1}{2}$ pound more of weight than he did the previous month. What was the weight of Captain Marbles, in pounds, at the end of April?

3.

4. A clock displaying a time of 2 o'clock is shown below. At how many times in the five hour period between 2 o'clock and 7 o'clock do the hour hand and minute hand of the clock form a right angle?

4.



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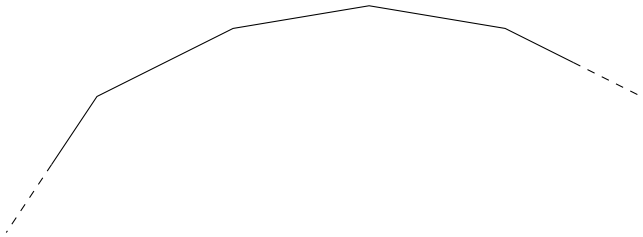
Problems 5 & 6

5. A palindrome is a number that is unchanged when the order of its digits is reversed. For example, the numbers 1551 and 515 are both palindromes. Mala chose a three-digit positive whole number that was a palindrome, and Nikki chose a four-digit positive whole number that was palindrome. The difference between Nikki's number and Mala's number was 544. What was Mala's number?

5.

6. A convex polygon has exactly four interior angles that are obtuse. What is the greatest possible number of sides for the polygon?

6.



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Problems 7 & 8

7. What is the only four-digit perfect square with a thousands digit that is greater than or equal to the hundreds digit, a hundreds digit that is greater than or equal to the tens digit, a tens digit that is greater than or equal to the units digit, and a units digit that is greater than zero?

7.

8. When performing addition, a carry is required when the sum of the digits in the same place in each number is greater than 9. For example, when adding 15 and 16, since 5 and 6 are in the units place and $5 + 6 = 11$, a carry to the tens place is required when adding 15 and 16. How many different pairs of consecutive three-digit positive integers require no carry when adding their values? For example, 123 and 124 are one such pair, since $4 + 4 = 8$, $2 + 2 = 4$, and $1 + 1 = 2$, and $123 + 124 = 247$.

8.

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School or Team

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6.	7.	8.	9.	10.

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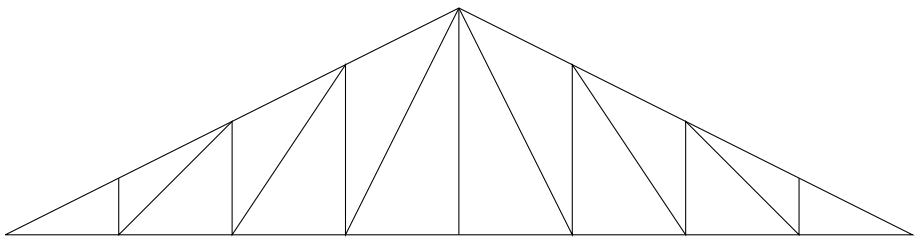
1. Mikey took a road trip with his family for Spring Break. Mikey and his family drove from their home in Houston to visit Mikey's aunt in Dallas, a distance of 246 miles. From Dallas, Mikey and his family drove to Little Rock to visit Mikey's grandmother, a distance of 324 miles. From Little Rock, Mikey and his family drove home to Houston. But Mikey and his family took a more direct route home, and the distance traveled was 131 miles shorter than the total distance they traveled from Houston to Dallas and then Dallas to Little Rock. How many total miles did Mikey and his family travel on their Spring Break road trip?

2. Wright wrote all of the whole numbers from 1 through 50. How many of the numbers Wright wrote contained either the digit 2 or the digit 3?

3. In the addition problem below, each letter represents a different digit. What is the greatest possible sum of the values of the digits represented by A, B, and C?

$$\begin{array}{r} A\ B \\ +\ A\ B \\ \hline C\ A \end{array}$$

4. The figure below shows a triangle partitioned into a number of smaller triangles. How many different triangles can be drawn by tracing some or all of the figure?



5. In the country of Primeland, coins have values of either 23 cents or 29 cents. What is the greatest whole number of cents less than 200 cents that a citizen of Primeland could make using an unlimited supply of these coins?

6. Elizabeth and her mother have the same birthday. The age of Elizabeth's mother is four times the age of Elizabeth. Four years ago, the age of Elizabeth's mother was seven times the age of Elizabeth. In how many years will the age of Elizabeth's mother be twice the age of Elizabeth?

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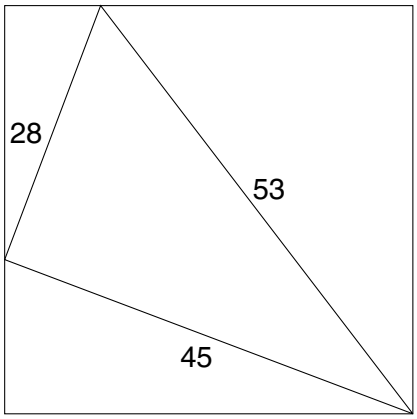
7. Rhonda rolled two standard dice. She looked at the result and told Samuel that the sum of the numbers shown on the dice was at least 8. With this information, Samuel calculated the probability that at least one of the dice showed a 4. What probability did Samuel calculate? Express your answer as a common fraction.

8. Jalen and Jason each picked a positive whole number less than 50. The greatest common divisor of Jalen's number and Jason's number was 6. How many different pairs of numbers could Jalen and Jason have picked? For example, Jalen picking 12 and Jason picking 18 is one such pair, and Jalen picking 18 and Jason picking 12 is another such pair.

9. As shown below, a triangle with sides of length 45, 28, and 53 shares one of its vertices with the vertex of a square, and the other two vertices of the triangle lie on sides of the square. To the nearest whole number, what is the area of the square?



10. In Pascal's triangle, the value of a number is the sum of the number above it just to the left and the number above it just to the right. The first few rows of Pascal's triangle are shown below. The top row is row 0. On what row number of Pascal's Triangle do three consecutive numbers appear that are in the ratio 2 : 3 : 4?

Row 0						1					
Row 1					1		1				
Row 2				1		2		1			
Row 3			1		3		3		1		
Row 4		1		4		6		4		1	
Row 5	1		5		10		10		5		1

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Answers 12227

Sprint Round

1. E
2. C
3. D
4. D
5. B
6. B
7. D
8. D
9. B
10. C
11. D
12. C
13. A
14. E
15. A
16. B
17. A
18. B
19. E
20. B
21. D
22. B
23. E
24. C
25. A
26. D
27. D
28. A
29. B
30. B

Target Round

1. 12
2. 847
3. 11
4. 9
5. 787
6. 7
7. 7744
8. 124

Team Round

1. 1009
2. 26
3. 20
4. 30
5. 197
6. 16
7. $\frac{1}{3}$
8. 43
9. 1772
10. 34

Number Sense

1. 199
2. 811
3. 108
4. 1
5. 206
6. 2
7. 159
8. 119
9. 23
10. $[7154, 7906]$
11. 1034
12. 256
13. 7
14. 90
15. 840
16. 290
17. 1710
18. 1351
19. 80
20. $[62198, 68744]$
21. 4
22. 1425
23. $\frac{7}{12}$
24. 33
25. 141
26. $\frac{3}{7}$
27. 27
28. $\frac{1}{10}$
29. 961
30. $[83799, 92619]$
31. 2024
32. 108
33. 28
34. 2880
35. 62
36. 364
37. 9025
38. 75
39. 4059
40. $[265313, 293239]$
41. 1325
42. 163
43. 2349
44. 360
45. 5
46. 5151
47. 121
48. 16
49. 1024
50. $[3664283, 4049995]$
51. 12285
52. 90
53. 110101
54. 7104
55. 52
56. 83
57. 21
58. 7469
59. 150
60. $[3895, 4304]$
61. 5625
62. $19\frac{4}{9}$
63. 20
64. $\frac{17}{30}$
65. 1728
66. 900
67. 8736
68. 244
69. $\frac{2}{7}$
70. $[118643, 131131]$
71. 1060899
72. 2
73. 42240
74. 128
75. 243
76. 8
77. 72
78. 12
79. 1331
80. $[7520, 8311]$

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Solutions 12227

Sprint Round Solutions

1. The total cost of 3 packages of gum is $3 \times \$2 = \$6$. When paying with a \$10 bill, the number of dollars received in return is $\$10 - \$6 = \boxed{\$4}$.

2. Each of the sums includes three whole numbers. If the sum of three whole numbers is odd, then either all three numbers must be odd, or two numbers must be even and the third must be odd. There is no answer choice where all three numbers are odd, and the only answer choice where two numbers are even and the third is odd is $\boxed{12 + 17 + 18}$.

3. The total number of oranges was $2 \times 12 = 24$, the total number of bananas was $3 \times 7 + 6 = 27$, and the total number of apples was $2 \times 10 = 20$. Altogether, the total number of oranges, bananas, and apples Kaitlyn purchased was $24 + 27 + 20 = \boxed{71}$.

4. The area of the square is $6 \times 6 = 36$. Since the rectangle has the same area, the length of the rectangle is $36 \div 4 = 9$. The perimeter of the rectangle is $9 + 9 + 4 + 4 = \boxed{26}$.

5. Before the doubling, the result was $44 \div 2 = 22$. Before subtracting 5, the result was $22 + 5 = 27$. And before the tripling, Linus' original number was $27 \div 3 = \boxed{9}$.

6. Evan's expression is equivalent to $(222 + 333 + 444) \times 17$, or 999×17. This is $17 \times 1000 - 17 \times 1$, or $17000 - 17 = 16983$, and $1 + 6 + 9 + 8 + 3 = \boxed{27}$.

7. The days of the month that were Saturdays were 2, 9, 16, 23, and 30, for a total of 5 Saturdays. The days of the month that were Sundays were 3, 10, 17, and 24, for a total of 4 Sundays. Altogether, the total number of days that were either Saturdays or Sundays was $4 + 5 = \boxed{9}$.

8. Since $6 = 3 \times 2$ and $9 = 3 \times 3$, for a number to be divisible by 9 and not divisible by 6, the number must be an odd multiple of 9. $200 \div 9$ is a little more than 22, so there are 22 multiples of 9 less than 200. Half of these are odd, so the count of numbers less than 200 that are divisible by 9 but not 6 is $22 \div 2 = \boxed{11}$.

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9. There are two possibilities. The first possibility is that the number is between 21 and 30. Then the distance from 21 to 30, which is $30 - 21 = 9$, is split into two segments by Chip's number, with one segment twice as long as the other. Therefore the distance from Chip's number to 21 must be $9 \div 3 = 3$, which means Chip's number is $21 + 3 = 24$. The second possibility is that the number is less than 21. The distance of this number from 30 is always 9 more than the distance from 21, so for the distance from 30 to be twice the distance from 21, the distance from 21 must be 9, and Chip's number must be $21 - 9 = 12$. The sum of the two possible values for Chip's number is $12 + 24 = \boxed{36}$.

10. The sum of $\frac{1}{5}$ and $\frac{1}{4}$ is $\frac{4}{20} + \frac{5}{20} = \frac{9}{20}$, so the value of the third number is $\frac{1}{2} - \frac{9}{20}$, or $\frac{10}{20} - \frac{9}{20} = \frac{1}{20}$. The second greatest of $\frac{1}{4}$, $\frac{1}{5}$, and $\frac{1}{20}$ is $\frac{1}{5}$, and the least is $\frac{1}{20}$. The value of the second greatest minus the least is $\frac{1}{5} - \frac{1}{20} = \frac{4}{20} - \frac{1}{20} = \boxed{\frac{3}{20}}$.

11. Let Optimus Prime be represented by O, Bumblebee by B, Megatron by M, and Starscream by S. Consider where Bumblebee is placed. If Bumblebee is in the leftmost position, then Optimus must be next, so the only possibilities are *BOMS* and *BOSM*, for 2 possibilities. Similarly, if Bumblebee is in the rightmost position, the only possibilities are *SMOB* and *MSOB*, for 2 possibilities. If Bumblebee is in the second leftmost position, the only possibilities are *OBSM* and *SBOM*, for 2 possibilities. Similarly, if Bumblebee is in the second rightmost position, the only possibilities are *MOBS* and *MSBO*, for 2 possibilities. Altogether, the total number of possible arrangements is $2 + 2 + 2 + 2 = \boxed{8}$.

12. The dimensions of the large rectangle are $4 + 6 = 10$ and $8 + 3 = 11$, so the area of the large rectangle is $10 \cdot 11 = 110$. The unshaded area of the large rectangle is two small right triangles, each with legs of length 3 and 4, having a total area of $3 \cdot 4 = 12$, and two large right triangles, each with legs of length 6 and 8, having a total area of $6 \cdot 8 = 48$. The area of the shaded rectangle is $110 - 12 - 48 = \boxed{50}$.

13. A ratio of $1 : 1$ means the number of fifth grade was equal to the number of sixth grade students. Because 4 fifth grade and 1 sixth grade student joined the team to make the ratio $1 : 1$, prior to these students joining there must have been 3 more sixth grade students than fifth grade students. Two numbers that are in the ratio of $5 : 6$ with a difference of 3 are 15 and 18. After the students join, the number of fifth and sixth grade students is $15 + 4$ and $18 + 1$, for a total of $19 + 19 = \boxed{38}$.

14. In the swimming race, Andrew finished in first place, and Brandon could not have finished in second place, since he finished in second place in the running race and did not finish in the same place in more than one race. Therefore Clint finished second place in the swimming race and Brandon finished in third place in the swimming race. In the cycling race, Brandon could not finish in third place, since he finished in third place in the swimming race, and he could not finish in second place, since he finished in second place in the running race. Therefore Brandon finished in first place in the cycling race. Clint could not finish in second place in the cycling race since he finished in second place in the swimming race. Therefore Clint finished in third place in the cycling race, and Andrew finished in second place in the cycling race, so the order of finish was $\boxed{B, A, C}$.

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15. The value of X is $8 - 5 = 3$. The value of the square with 23 is equal to the sum of the 2 and the 3 from the bottom row, plus twice the value of the unknown square from the bottom row, as this number is included in the sum for each of the squares immediately below the 23. Therefore the value of the unknown square on the bottom row is $(23 - 2 - 3) \div 2 = 9$. The value of the middle square in the next-to-bottom row is $9 + 3 = 12$, and the value of the rightmost square in the next-to-top row is $12 + 8 = 20$. Finally, the value of the top square is $23 + 20 = \boxed{43}$.

16. There were 3 games in the first match, so the sum of Emily's digs for the first match is $11 \cdot 3 = 33$. In the second match, Emily had an additional $9 + 14 + 16 = 39$ digs, so her total number of digs for the two matches was $33 + 39 = 72$. She played a total of 6 games, so her average number of digs per game is $\frac{72}{6} = \boxed{12}$.

17. The perpendicular segment is the longer leg of a right triangle with a hypotenuse of length 13 and a shorter leg of length 5. By the Pythagorean Theorem, the length of this perpendicular segment is $\sqrt{13^2 - 5^2} = 12$. The side of the large triangle with an unknown length is the hypotenuse of a right triangle with legs of length 12 and 9. By the Pythagorean Theorem, the length of this side is $\sqrt{12^2 + 9^2} = 15$, and the perimeter of the large triangle is $13 + 5 + 9 + 15 = \boxed{42}$.

18. Because $\sqrt{200}$ is a little less than 14, the only primes that can possibly divide a composite number less than 200 are 2, 3, 5, 7, 11, and 13. The multiples of 2 are all even numbers, so those cannot be prime. The multiples of 5 include all numbers with a units digit of 5, so those cannot be prime. The multiples of 3 are all numbers with digits that sum to a multiple of 3. This condition eliminates 153, 159, 171, 177, 183, and 189. Not being a multiple of 7 eliminates 161. Not being a multiple of 11 eliminates 187. Not being a multiple of 13 eliminates 169. This leaves 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, and 199, for a total of $\boxed{11}$.

19. Devan runs twice as fast as he walks, so walking any distance takes twice as much time as running. Had Devan walked the last $\frac{1}{4}$ mile, it would have taken him $2 \cdot 2\frac{1}{2} = 5$ minutes. Since Devan can walk $\frac{1}{4}$ mile in 5 minutes, he can walk 1 mile in $5 \cdot 4 = 20$ minutes. Thus the length of his trip to school, in minutes, was $2\frac{1}{2} + 20 = \boxed{22\frac{1}{2}}$.

20. Since there is a border on all four edges of the the mosaic, there are $27 + 1 = 28$ borders of 0.25 inches added to the width, for a total horizontal width of $27 + 28 \cdot 0.25 = 34$. Similarly, there are $17 + 1 = 18$ borders of 0.25 inches added to the height, for a total height of $17 + 18 \times 0.25 = 21.5$. The total area covered by the mosaic, in square inches, is 34×21.5, which is $17 \times 43 = \boxed{731}$.

21. The expression $4ab + 2a + 2b + 1$ can be factored as $(2a + 1)(2b + 1)$, so $126 \diamondsuit 128 = (2 \cdot 126 + 1)(2 \cdot 128 + 1)$, or $253 \cdot 257$. This is equivalent to $250^2 + 3 \cdot 250 + 7 \cdot 250 + 3 \cdot 7 = 250 \cdot 260 + 21$, or $\boxed{65021}$.

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22. The area of the base of the pyramid is $16^2 = 256$. Each triangular face of the pyramid is an isosceles triangle with two sides of length 17 and one side of length 16. Dropping an altitude from the vertex where the two equal sides of length 17 meet to the side of length 16 partitions the isosceles triangle into two congruent right triangles, each with a hypotenuse of length 17 and a leg of length $\frac{16}{2} = 8$. By the Pythagorean Theorem, the length of the second leg, which is also the length of the altitude of the isosceles triangle, is $\sqrt{17^2 - 8^2} = 15$. Therefore each of the four triangular faces of the pyramid has area $\frac{1}{2} \cdot 16 \cdot 15 = 120$, and the total surface area of the pyramid is $120 \cdot 4 + 256 = \boxed{736}$.

23. Let X be the location of a K, N, or B, and _ be the location of any other letter. Any arrangement not having a K next to a N, a K next to a B, and a N next to a B must be of the form $X_X_X_$, $X_X__X$, $X__X_X$, or $_X_X_X$. Altogether, this is 4 possible forms. Within each form, there are $3! = 6$ ways to position the K, N, and B, and $3! = 6$ ways to position the remaining letters. Therefore the total number of possible arrangements is $4 \cdot 6 \cdot 6 = \boxed{144}$.

24. The number 512 is 8^3, or the cube of 11_7. The cube of 11_7 is $(110_7 + 11_7) \cdot 11_7$, or $121_7 \cdot 11_7$. This is $1210_7 + 121_7 = 1331_7$. Huey's number in base 9 was 1331_9. This could be converted directly, but it is also the cube of 11_9, or 10. Therefore Huey's number was $10^3 = \boxed{1000}$.

25. The number 10! is $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$. As a prime factorization, this is $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. For a divisor to be a perfect square, the divisor must have a prime factorization with exponents that are all even. Therefore it must have a prime factor of 2 with an exponent of 0, 2, 4, 6, or 8, for 5 possibilities. It must have a prime factor of 3 with an exponent of 0, 2, or 4, for 3 possibilities, and it must have an prime factor of 5 with an exponent of 0 or 2, for 2 possibilities. The total number of divisors of 10! that are perfect squares is $5 \cdot 3 \cdot 2 = \boxed{30}$.

26. Each time Dan passes Dave, the father and son have, together, completed 1 lap around the track. In 65 seconds, Dan completes $\frac{65}{90} = \frac{13}{18}$ of a lap. Therefore Dave must complete $1 - \frac{13}{18} = \frac{5}{18}$ of a lap in 65 seconds. If Dave completes $\frac{5}{18}$ of a lap in 65 seconds, then his time to complete 1 lap is $65 \div \frac{5}{18} = 13 \cdot 18$, which is $\boxed{234}$.

27. The larger hexagon is comprised of six equilateral triangles with altitudes of length 6. The side lengths of each of these equilateral triangles is $\frac{6}{\sqrt{3}} \cdot 2 = 4\sqrt{3}$. The smaller hexagon is comprised of six equilateral triangles with sides of length 6. The area of an equilateral triangle with side length s is $\frac{s^2 \cdot \sqrt{3}}{4}$. The area of the larger hexagon is $6 \cdot \frac{(4\sqrt{3})^2 \cdot \sqrt{3}}{4}$, and the area of the smaller hexagon is $6 \cdot \frac{6^2 \cdot \sqrt{3}}{4}$. The area region within exactly one of the hexagons is the difference between these two values, or $\frac{6}{4} \cdot \sqrt{3} \cdot \left((4\sqrt{3})^2 - 6^2\right) = \frac{3}{2} \cdot \sqrt{3} \cdot (48 - 36)$, which is $\boxed{18\sqrt{3}}$.

28. Since the numbers in the list form an arithmetic sequence and the first number is 3, the remaining six numbers can be expressed as $3+k$, $3+2k$, $3+3k$, $3+4k$, $3+5k$, and $3+6k$. The sum of the first three numbers is $9 + 3k$, so the average is $3 + k$, and the sum of the last four numbers is $12 + 18k$, so the average is $3 + \frac{9}{2}k$. Because the latter average is three times the former average, $3 \cdot (3 + k) = 3 + \frac{9}{2}k$, and $6 = \frac{3}{2}k$, so $k = 4$. Then the fourth number is $3 + 3 \cdot 4 = 15$ and the fifth number is $15 + 4 = 19$, and their sum is $15 + 19 = \boxed{34}$.

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29. Multiplying the entire expression by 2 yields $2 \cdot \sqrt{1.5 + 0.5\sqrt{5}} - \sqrt{5} + 1$, or $\sqrt{6 + 2\sqrt{5}} - \sqrt{5} + 1$. Considering only $\sqrt{6 + 2\sqrt{5}}$, suppose this is equal to $\sqrt{a} + \sqrt{b}$ for some positive whole numbers a and b. Then, squaring, $6 + 2\sqrt{5} = a + b + 2\sqrt{ab}$. The right side is equivalent to the left side if either of a or b equals 5 and the other is 1, so $\sqrt{6 + 2\sqrt{5}} = \sqrt{1} + \sqrt{5}$, or $1 + \sqrt{5}$. Substuting back into the original expression yields $1 + \sqrt{5} - \sqrt{5} + 1 = 2$. Since the original expression was multiplied by 2, this is twice the original value, which is $\boxed{1}$.

30. A line is unique if the ratio of the vertical distance from the lower left dot to the horizontal distance from the lower left dot is unique. The ratio is unique if, when expressed as a fraction, it cannot be reduced. The vertical distance may be any value from 0 through 9, and the horizontal distance may also be any value from 0 through 9. Consider the lines with such a positive ratio less than 1. There is 1 such ratio with a denominator of 2, 2 such ratios with a denominator of 3, and an additional 2 such ratios with a denominator of 4. There are 4 with a denominator of 5, 2 with a denominator of 6, and 6 with a denominator of 7. Finally, there are 4 with a denominator of 8 and 6 with a denominator of 9. This is a total of $1 + 2 + 2 + 4 + 2 + 6 + 4 + 6 = 27$ lines. By symmetry, there are an equal number of lines with such a ratio greater than 1. Additionally, there is a horizontal line, a vertical line, and a line along the diagonal of the grid. Therefore the total number of unique lines is $27 \cdot 2 + 1 + 1 + 1 = \boxed{57}$.

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Target Round Solutions

1. There are 9 cookies in a packet, 2 packets per column, and 2 columns per box. The total number of cookies in the box is $2 \times 2 \times 9 = 36$. If Maddy ate 3 cookies per day, then the number of days it took her eat all of the cookies in the box was $36 \div 3 = \boxed{12}$.

2. Any number that is a multiple of both 7 and 11 must be a multiple of $7 \times 11 = 77$. Any number that has a remainder of 2 when divided by 13 is $13 - 2 = 11$ less than a multiple of 13. Because 77 is 1 less than a multiple of 13, $77 \times 11 = 847$ is 11 less than a multiple of 13, so the number that Thomas chose was $\boxed{847}$.

3. At end of January, Captain Marbles weighed $4 + 1 = 5$ pounds. In February, Captain Marbles gained $1 + \frac{1}{2} = 1\frac{1}{2}$ pounds, so his weight at the end of February was $5 + 1\frac{1}{2} = 6\frac{1}{2}$ pounds. In March, Captain Marbles gained $1\frac{1}{2} + \frac{1}{2} = 2$ pounds, so his weight at the end of March was $6\frac{1}{2} + 2 = 8\frac{1}{2}$ pounds. In April, Captain Marbles gained $2 + \frac{1}{2} = 2\frac{1}{2}$ pounds, so his weight at the end of April, in pounds, was $8\frac{1}{2} + 2\frac{1}{2} = \boxed{11}$.

4. The hour hand and minute hand form a right angle when the minute hand is either 90° counterclockwise from the hour hand, or the minute hand is 90° clockwise from the hour hand. In the 2 o'clock hour, the minute hand will never be 90° counterclockwise from the hour hand, but it will be 90° clockwise from the hour hand, for 1 right angle formed. This will occur at a time shortly after $2:25$. In the 3 o'clock hour, the minute hand will be 90° behind the hour hand at exactly 3 o'clock, and be 90° ahead later in the hour, at a time shortly after $3:30$, for 2 right angles. In the 4 o'clock, 5 o'clock, and 6 o'clock hours, 2 right angles are formed for each hour. The total number of right angles formed is $1 + 2 \cdot 4 = \boxed{9}$.

5. For the two palindromes to differ by less than 1000, the four-digit palindrome must be between 1000 and 2000, so both the thousands digit and units digit of the four-digit palindrome are 1. If the four-digit palindrome has a units digit of 1, and the difference with the three-digit palindrome is 544, then the three-digit palindrome must have a units digit of $11 - 4 = 7$. This is also the hundreds digit of the three-digit palindrome. Thus the four-digit palindrome is of the form $1AA1$, where A is a digit, and the three-digit palindrome is of the form $7B7$, where B is a digit. Since $544 + 707$ is greater than 1221, and $544 + 797$ is less than 1441, the four-digit palindrome must be 1331, so the value of the three-digit palindrome, Mala's number, is $1331 - 544 = \boxed{787}$.

6. The sum of the exterior angles of any convex polygon is 360°. If an interior angle at a vertex is not obtuse, then it is acute or right, and the exterior angle at that vertex must be greater than or equal to 90°. If 4 exterior angles are greater than or equal to 90°, then, after including the acute exterior angles at the four given vertices, the sum of the exterior angles would be greater than 360°. Therefore the greatest number of exterior angles that can be greater than or equal to 90°, and therefore interior angles less than or equal to 90°, is 3, so there are at most 3 other vertices, and the greatest possible value for the number of sides is $4 + 3 = \boxed{7}$.

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7. Since $32^2 = 1024$ and $99^2 = 9801$, any four-digit perfect square must be the square of a number greater than or equal to 32 and less than or equal to 99. Given the conditions, the units digit should be as small as possible and the thousands digit should be as large as possible. The only way to have a units digit of 1 is to square a number with a units digit of 1 or 9. Checking 39, 41, 49, 51, 59, 61, 69, 71, 79, 81, 89, 91, and 99, none meet the given conditions. The next smallest possible value of the units digit of a square is 4, which can only occur when squaring a number with a units digit of 2 or 8. However, if the units digit is a 4, then the thousands digit must be at least 4, eliminating squares of numbers less than $\sqrt{4000} \approx 63$. Checking 68, 72, 78, 82, and 88, the value of 88^2 is 7744. Therefore, the only four-digit perfect square with digits that meet the stated conditions must be $\boxed{7744}$.

8. The hundreds digit of the smaller number must be 1, 2, 3, or 4, since adding any two numbers greater than or equal to 500 requires a carry. Similarly, the tens digit of the smaller number must be 0, 1, 2, 3, or 4, or the tens and units digits of the smaller number together must be 99. Finally, the units digit of the smaller number must be 0, 1, 2, 3, 4, or 9. Counting the possibilities for the smaller number when no digit is a 9, there are 4 possibilities for the hundreds digit, 5 for the tens digit, and 5 for the units digit, for $4 \cdot 5 \cdot 5 = 100$ possibilities. Counting the possibilities for the smaller number when the tens and units digits are 99, there are 4 possibilities, namely 199, 299, 399, and 499. Counting the possibilities for the smaller number when only the units digit is a 9, there are 4 possibilities for the hundreds digit and 5 possibilities for the tens digit, for 20 possibilities. Therefore the total number of possible pairs is $100 + 4 + 20 = \boxed{124}$.

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Team Round Solutions

1. The distance traveled from Houston to Dallas and then Dallas to Little Rock was $246+324 = 570$ miles. The distance from Little Rock to Houston, however, was $570 - 131 = 439$ miles. The total distance traveled, in miles, was $570 + 439 = \boxed{1009}$.

2. The digit 2 occurred in 10 numbers from 20 through 29, and also occurred in the numbers 2, 12, 32, and 42, for a total of 14. Because 32 was already counted, the digit 3 occurred in 9 uncounted numbers from 30 through 39, and also occurred in the uncounted numbers 3, 13, and 43, for a total of 12. Altogether, the total number of numbers that contain either the digit 2 or the digit 3 is $14 + 12 = \boxed{26}$.

3. Because $B + B$ sums to A in the units digit, the value of A must be even. But the sum of A and A in the tens digit must be less than 10, as there is no hundreds digit, so the greatest possible value of A is 4. This will also maximize C, since C is either $A + A$ or, if there is a carry from the units digit, $A + A + 1$. There are two possible values of B that produce a value of A that is 4. The first is $B = 2$. This yields $42 + 42 = 84$. The second is $B = 7$. This yields $47 + 47 = 94$. The second produces the greatest possible sum of A, B, and C, which is $4 + 7 + 9 = \boxed{20}$.

4. The entire triangle is partitioned into 14 smaller triangles. Using exactly 2 of the small triangles, there are 3 triangles that exist, one on the left side of the figure, one on the right side of the figure, and one in the middle of the figure. Using exactly 3, 4, 5, 6, 7, or 8 of the triangles, there are 2 of each, one on the left side of the figure and one on the right side of the figure. This accounts for $6 \cdot 2 = 12$ triangles. There is also 1 largest triangle formed by all of the small triangles. Altogether, the total number of triangles is $14 + 3 + 12 + 1 = \boxed{30}$.

5. Because $29 \times 7 = 203$, the greatest number of 29 cent coins that could be used is 6. If 6 coins worth 29 cents are used, they have a value of $6 \times 29 = 174$, and 1 coin worth 23 cents could be added to bring the total to 197 cents. If only 5 are used, then the value is $29 \times 5 = 145$, and only 2 of the coins worth 23 cents could be added to bring the total to $145 + 2 \times 23 = 191$ cents. If only 4 are used, then the value is $29 \times 4 = 116$, and only 3 of the coins worth 23 cents could be added to bring the total to $116 + 3 \times 23 = 185$. If only 3 are used, then the value is $29 \times 3 = 87$, and only 4 of the coins worth 23 cents could be added to bring the total to $87 + 4 \times 23 = 179$. If only 2 are used, then the value is $29 \times 2 = 58$, and only 6 of the coins worth 23 cents could be added to bring the total to $58 + 6 \times 23 = 196$. If only 1 is used, then the value is $29 \times 1 = 29$, and only 7 of the coins worth 23 cents could be added to bring the total to $29 + 7 \times 23 = 190$. Finally, if none are used, then the greatest total could be $23 \times 8 = 184$ cents. The greatest whole number of cents less than 200 cents that is possible using these coins is therefore $\boxed{197}$.

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6. Let Elizabeth's current age be E. Then Elizabeth's mother's current age is $4E$. Four years ago, Elizabeth's age was $E-4$, and Elizabeth's mother's age was $4E-4$. But since Elizabeth's mother's age was 7 times Elizabeth's age four years ago, $7 \cdot (E-4) = 4E-4$, or $3E = 24$, and $E = 8$. Then Elizabeth's mother's current age is $4 \cdot 8 = 32$. Since Elizabeth is always $32 - 8 = 24$ years younger than her mother, Elizabeth's mother will be twice the age of Elizabeth when Elizabeth is 24. The number of years from now that this will occur is $24 - 8 = \boxed{16}$.

7. When rolling two standard dice, the number of ways to have a sum of 8 is 5, a sum of 9 is 4, a sum of 10 is 3, a sum of 11 is 2, and a sum of 12 is 1. Therefore the dice show one of $5+4+3+2+1 = 15$ outcomes. Of these, the outcomes $(4,4)$, $(4,5)$, $(5,4)$, $(4,6)$ and $(6,4)$ include at least one 4. There are 5 such outcomes, for a probability of $\frac{5}{15} = \boxed{\frac{1}{3}}$.

8. If the lesser of the two numbers was 6, then the other number could be any multiple of 6 less than 50, for 8 possibilities. If the lesser of the two numbers was 12, then the other number could be any odd multiple of 6, for 3 possibilities. If the lesser of the two numbers was 18, then the other number could be any multiple of 6 other than 36, for 4 possibilities. If the lesser of the two numbers was 24, then the other number could be any odd multiple of 6, for 2 possibilities. If the lesser number was 30, then the other number could be any multiple of 6, for 3 possibilities. Finally, if the lesser number was 36, then the other number could be 42, for 1 possibility, and if the lesser number was 42, then the other number could be 48, for 1 possibility. Altogether, the number of possible pairs is $8+3+4+2+3+1+1 = 22$ pairs. However, Jalen could have either the greater or lesser number, except for the pair where both Jalen and Jason pick the number 6. Therefore the total number of different pairs is $22 \cdot 2 - 1 = \boxed{43}$.

9. Because $28^2 + 45^2 = 53^2$, the triangle is a right triangle. Additionally, the right triangle with a right angle at the lower left corner of the square and a hypotenuse of length 45 is similar to the right triangle with a right angle at the upper left corner of the square and a hypotenuse of length 28. Let the length of a side of the square be x. This is the longer leg of the right triangle with a hypotenuse of length 45. The longer leg of the right triangle with a hypotenuse of length 28 is $\frac{28}{45}x$. But this longer leg plus the shorter leg of the right triangle with the hypotenuse of length 45 are the side of the square, so the shorter leg of the right triangle with the hypotenuse of length 45 is $x - \frac{28}{45}x = \frac{17}{45}x$. By the Pythagorean Theorem, $\left(\frac{17}{45}x\right)^2 + x^2 = 45^2$. This is approximately $1.1427x^2 = 2025$. Then x^2, or the area of the square, is $\frac{2025}{1.1427}$, which to the nearest whole number is $\boxed{1772}$.

10. If the leftmost number of row n of Pascal's Triangle is called the $0th$ number in that row, then the kth number in the nth row is $\binom{n}{k}$. Let $\binom{n}{k}$ be the middle number of the three consecutive numbers in the ratio $2:3:4$. Then $\frac{3}{2} \cdot \binom{n}{k-1} = \binom{n}{k}$, and $\frac{4}{3} \cdot \binom{n}{k} = \binom{n}{k+1}$. The first equation becomes $\frac{3}{2} \cdot \frac{n!}{(k-1)! \cdot (n-(k-1))!} = \frac{n!}{k! \cdot (n-k)!}$, or $3 \cdot k! \cdot (n-k)! = 2 \cdot (k-1)! \cdot (n-(k-1))!$. This is $3 \cdot k = 2 \cdot (n-k+1)$, or $n = \frac{5}{2}k - 1$. The second equation becomes $\frac{n!}{k! \cdot (n-k)!} = \frac{3}{4} \cdot \frac{n!}{(k+1)! \cdot (n-(k+1))!}$, or $4 \cdot (k+1)! \cdot (n-(k+1))! = 3 \cdot k! \cdot (n-k)!$. This is $4 \cdot (k+1) = 3 \cdot (n-k)$. Substituting $n = \frac{5}{2}k - 1$ for n yields $4 \cdot (k+1) = 3 \cdot (\frac{5}{2}k - 1 - k)$, and $k = 14$, so n is $\frac{5}{2} \cdot 14 - 1 = \boxed{34}$.